

YUBO ZHAO

Software Engineering | Machine Learning | Reinforcement Learning | Robotics

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TECHNICAL SKILLS

Programming & Systems: Python, C, C++, Java, JavaScript, SQL, HTML/CSS, Git/GitHub, Linux, REST APIs, OOP, data structures & algorithms, IPC

ML / AI: PyTorch, scikit-learn, NumPy, pandas, OpenCV, MediaPipe, Gymnasium-style RL environments, PPO, CNNs, Vision Transformers, YOLO, transfer learning, cross-validation

Robotics & Tools: Java robotics programming, autonomous control, sensor integration, PID/IMU fundamentals, VS Code, Ollama/local LLM integration, speech recognition

EXPERIENCE & COMPETITIONS

Team Canada Member - International Olympiad in Artificial Intelligence (IOAI) | Competitive Artificial Intelligence 2025

- Selected to represent Team Canada at IOAI 2025, applying machine learning, deep learning, computer vision, NLP, data preparation, and model optimization under competition constraints.
- Designed and evaluated experimental ML pipelines with emphasis on generalization, performance analysis, and efficient iteration across advanced AI problem settings.

Robotics Team Member - FIRST Tech Challenge World Championship | FTC World Championship - 2nd Place Globally 2024

- Programmed and debugged Java-based robot software spanning autonomous routines, driver-control logic, sensor integration, and iterative performance tuning for world-championship competition.
- Integrated software with mechanical/electrical subsystems and competition strategy, contributing to a 2nd-place global finish through reliable autonomous and tele-operated execution.

Machine Learning / Quantitative Research Intern | Shaanxi Shun Yicheng Investment Management Co., Ltd. | Xi'an, China Jul-Aug 2025

- Built and evaluated financial ML workflows across preprocessing, feature engineering, predictive modeling, signal validation, and backtesting, emphasizing leakage prevention and out-of-sample generalization.
- Compared model behavior and risk characteristics using performance benchmarking and overfitting controls to support more defensible quantitative research decisions.

SELECTED PROJECTS

Warcraft III Reinforcement Learning Agent | Python | Lua | RL | PPO | Gymnasium | Game AI

- Engineered a custom reinforcement-learning environment for Warcraft III: Reforged, instrumenting the Lua runtime to expose real-time combat state and programmable unit controls to an external Python agent.
- Built bidirectional Python-Lua IPC and a synchronized action queue to work around game-sandbox/file-caching constraints, enabling movement, targeting, attacks, abilities, and streaming of HP, position, distance, targets, and combat state.
- Defined Gymnasium-style state, discrete action, and reward interfaces with a PPO-oriented training architecture for RTS micro, progressing from controlled 1v1 hero combat toward more complex tactical scenarios.

RoomRead - AI Behavioral Intelligence Engine | Next.js | TypeScript | OpenAI | Supabase | Exa | LLM Systems

- Built an AI intelligence engine that reconstructs how decision-makers evaluate companies, transforming fragmented interviews, essays, and transcripts into structured, source-grounded behavioral models.
- Designed an end-to-end evidence pipeline spanning discovery, extraction, semantic filtering, structured observations, contradiction tracking, confidence qualification, and deterministic pitch analysis with auditable source provenance.
- Developed a pitch red-team system that surfaces evaluator-specific objections, weak assumptions, and follow-up questions; hardened the research stack with versioned persistence, provider abstraction, provenance tracking, and 64-test validation.

Brain Tumor Detection System | Python | PyTorch | MiniMaxViT | Medical Imaging

- Developed an MRI classification pipeline around a compact vision-transformer architecture, testing preprocessing, augmentation, oversampling, class balancing, and validation strategies to improve robustness.
- Evaluated precision, recall, F1, confusion matrices, generalization, and potential dataset artifacts rather than relying on accuracy alone.

Jarvis - Local Multimodal AI Assistant | Python | Ollama | OpenCV | Speech | APIs

- Built a modular local assistant combining LLM inference, speech-to-text, text-to-speech, memory, external APIs, and real-time computer vision across voice, text, and camera input.

HONOURS

Euclid Mathematics Contest 2026 - Honour Roll, 86/100 | CSMC 2025 - Honour Roll, 48/60 | Sir Isaac Newton Physics Contest - Top 5% Globally

EDUCATION

University of Waterloo - Bachelor of Software Engineering | 1A Software Engineering | Waterloo, Ontario

Expected 2031